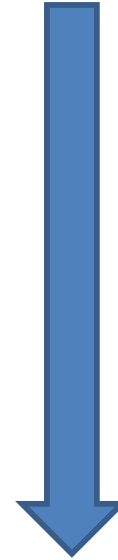


Operating System Structure

- General-purpose OS is very large program
- Various ways to structure
 - Simple structure – MS-DOS
 - More complex -- UNIX
 - Layered – an abstraction
 - Microkernel -Mach



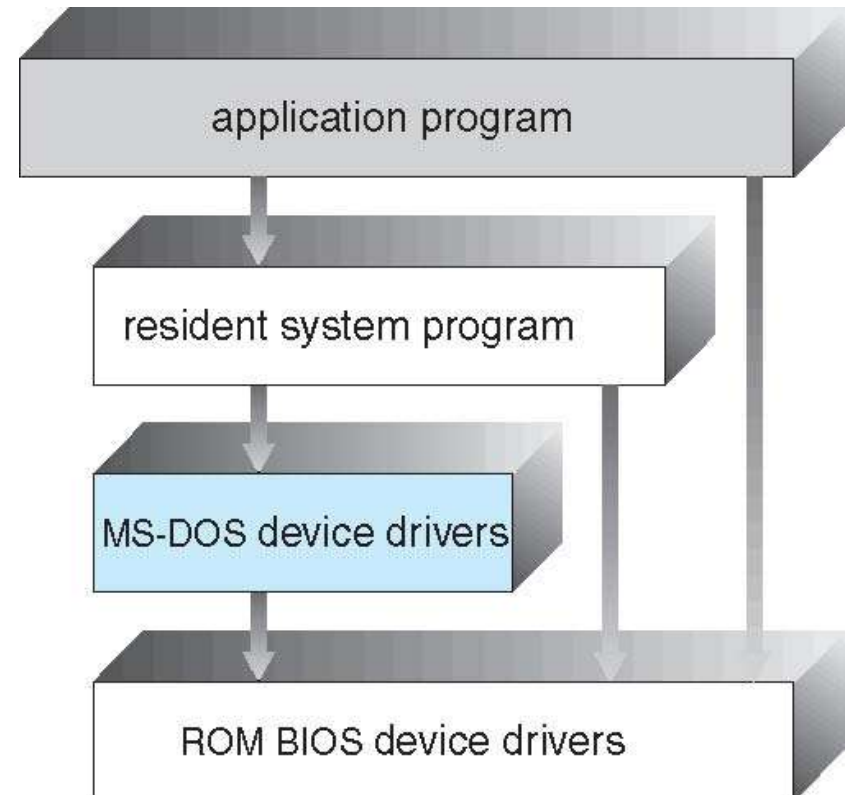
Simple Structure -- MS-DOS

- MS-DOS – written to provide **the most functionality in the least space**
 - Designed and **implemented by a few people who had no idea that it would become so popular**
 - **Written by Tim Paterson**



Simple Structure -- MS-DOS

- MS-DOS – written to provide the most functionality in the least space
 - **Because of the limited hardware it was build on,**
 - **it was Not divided into modules carefully**
 - Although MS-DOS has **some structure,**
 - **its interfaces and levels of functionality**
 - **are not well separated**



Simple Structure -- MS-DOS

- **Development?**
 - **Microsoft Disk Operating System**
 - OS for **x86-based PCS** mostly developed by Microsoft.
 - Collectively, MS-DOS,
 - **some operating systems attempting to be compatible with MS-DOS**, are sometimes referred to as "DOS" (which is also the generic acronym for disk operating system)

Simple Structure -- MS-DOS

- **Competitors?**
 - It was gradually **superseded by OS offering a graphical user interface (GUI),**
 - in various generations of the **graphical Microsoft Windows operating system.**

Simple Structure -- MS-DOS

- **Competitors?**
 - Although **MS-DOS** and **PC DOS** were initially developed in parallel by Microsoft and IBM,
 - the **two products diverged after twelve years, in 1993, with recognizable differences in compatibility, syntax, and capabilities.**

Simple Structure -- MS-DOS

- **Closure?**
 - MS-DOS went through **eight versions, until development ceased in 2000**
 - Versions
 - MS-DOS 1.x
 - MS-DOS 2.x
 - MS-DOS 3.x
 - MS-DOS 4.0 / MS-DOS 4.x
 - MS-DOS 5.x
 - MS-DOS 6.x
 - MS-DOS 7 (as part of Windows 9x)
 - Localized versions

Non Simple Structure -- UNIX

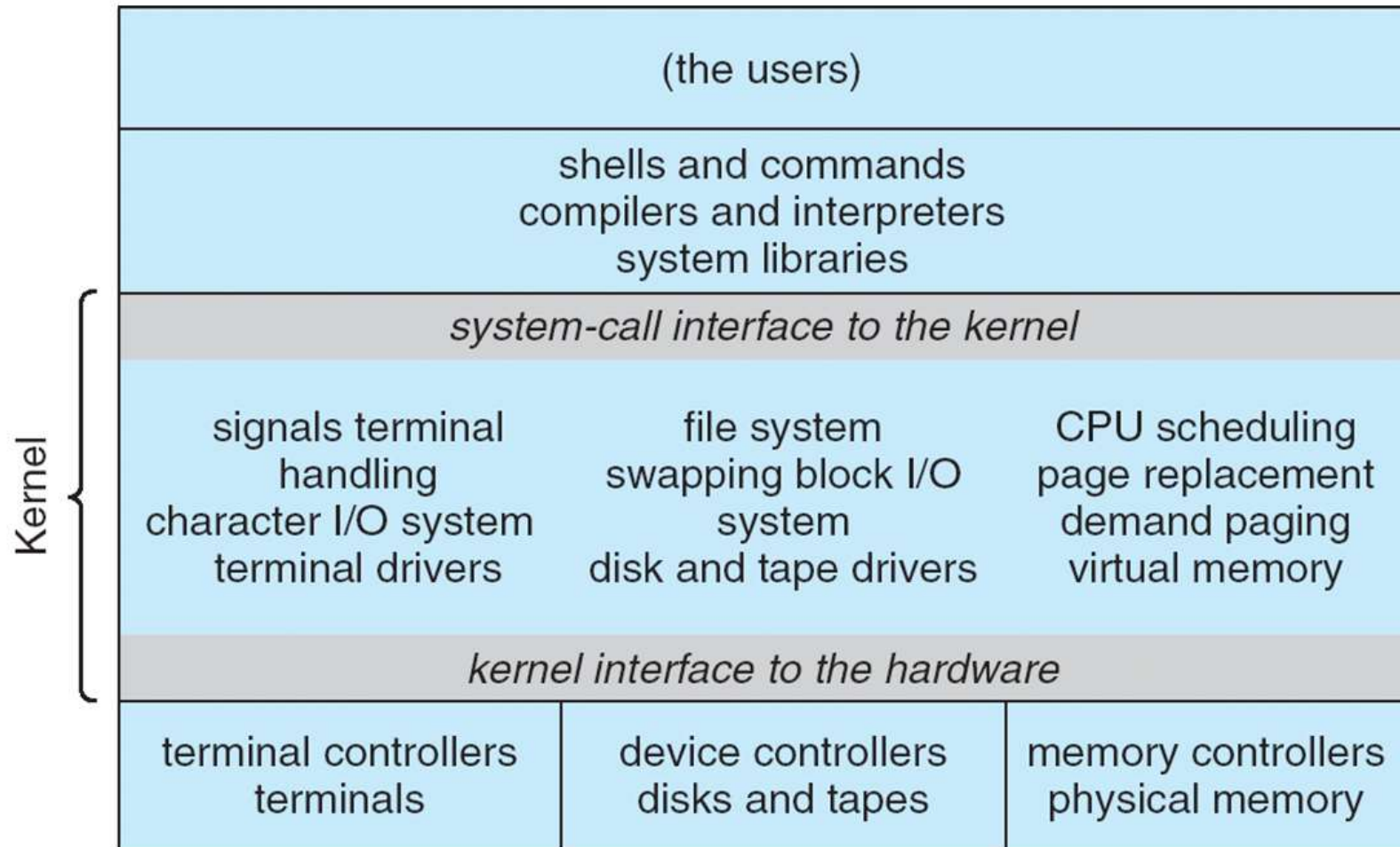
UNIX – **limited by hardware functionality**, the original UNIX operating system had limited structuring.

The UNIX OS consists of two separable parts

- **Systems programs**
- **The kernel**

Traditional UNIX System Structure

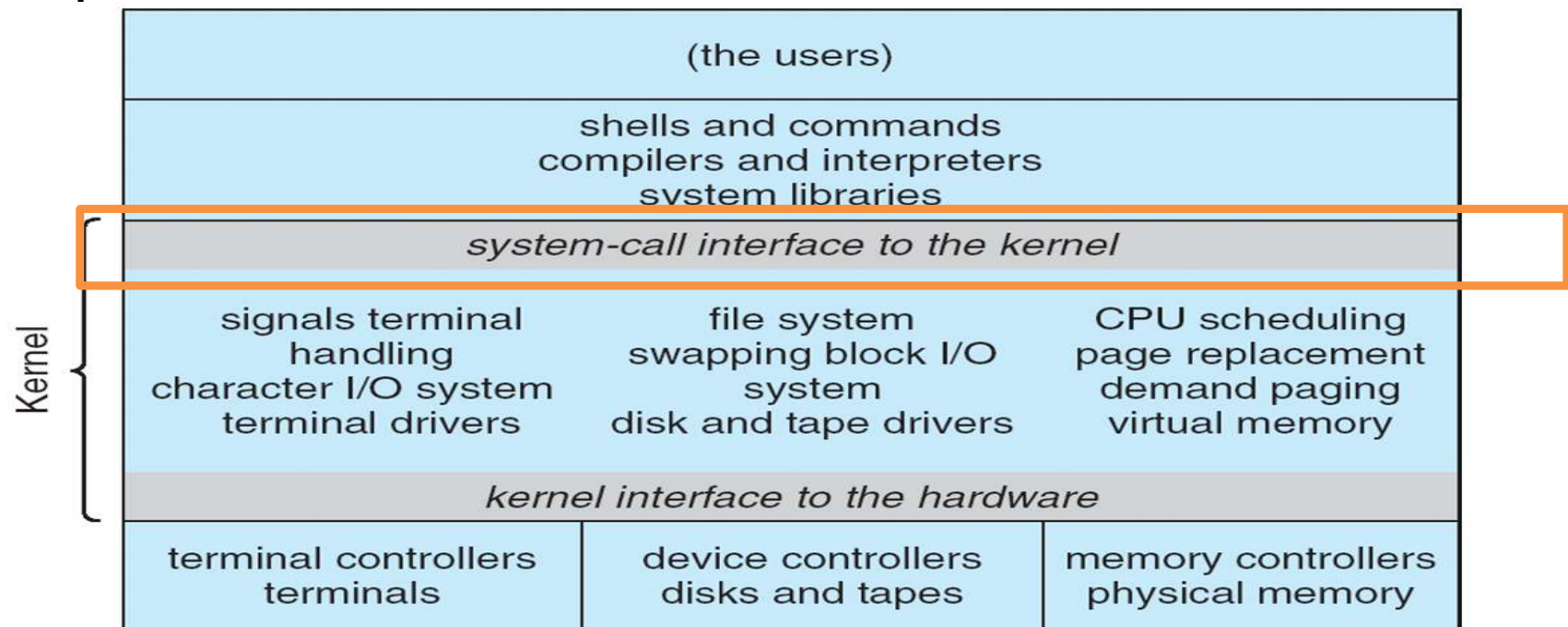
Beyond simple but not fully layered



Non Simple Structure -- UNIX

The kernel

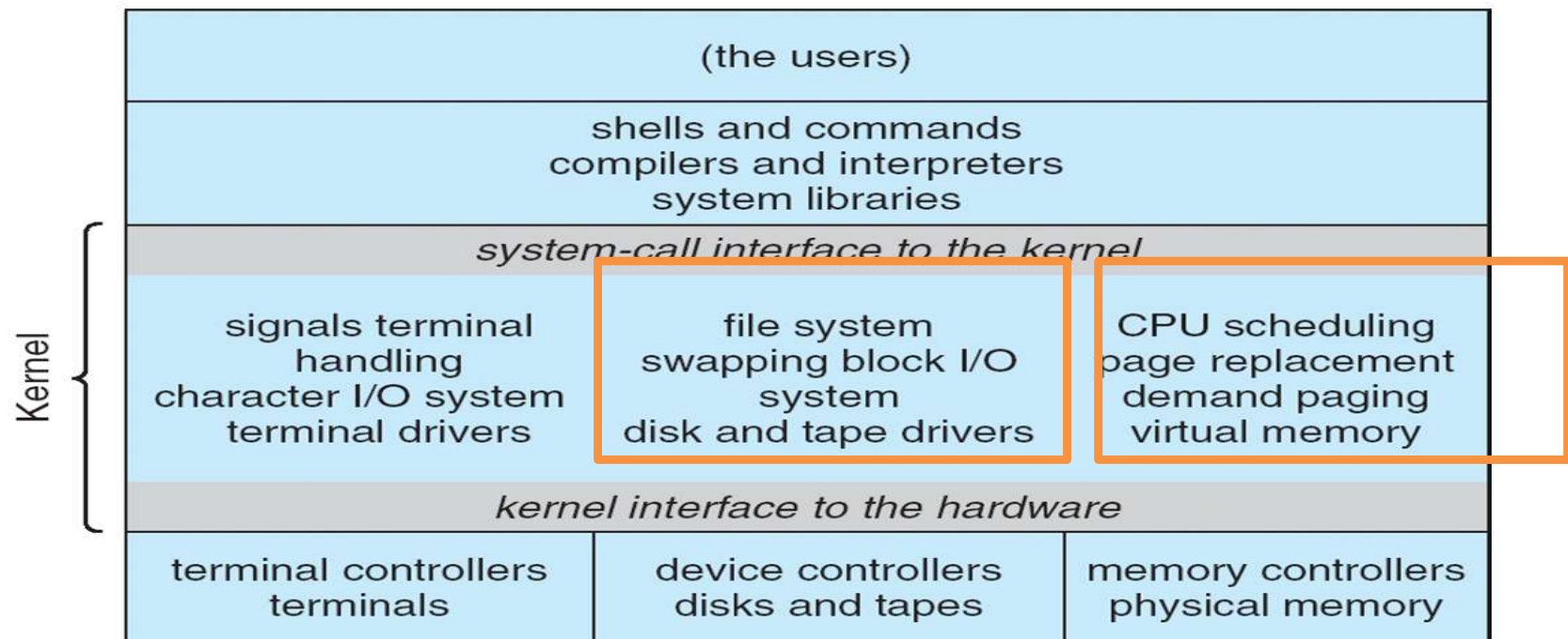
- Consists of everything **below the system-call interface and above the physical hardware**
- Further separated into series of interfaces and **device drivers which were added and expanded over the years** as UNIX evolved.



Non Simple Structure -- UNIX

The kernel

- Provides the **file system, CPU scheduling, memory management, and other functions; a large number of functions for one level**
- An **enormous amount of functionality to be combined into one level,**
- This makes **UNIX difficult to enhance,**
 - as **changes in one level could adversely affect other areas.**



UNIX System Structure

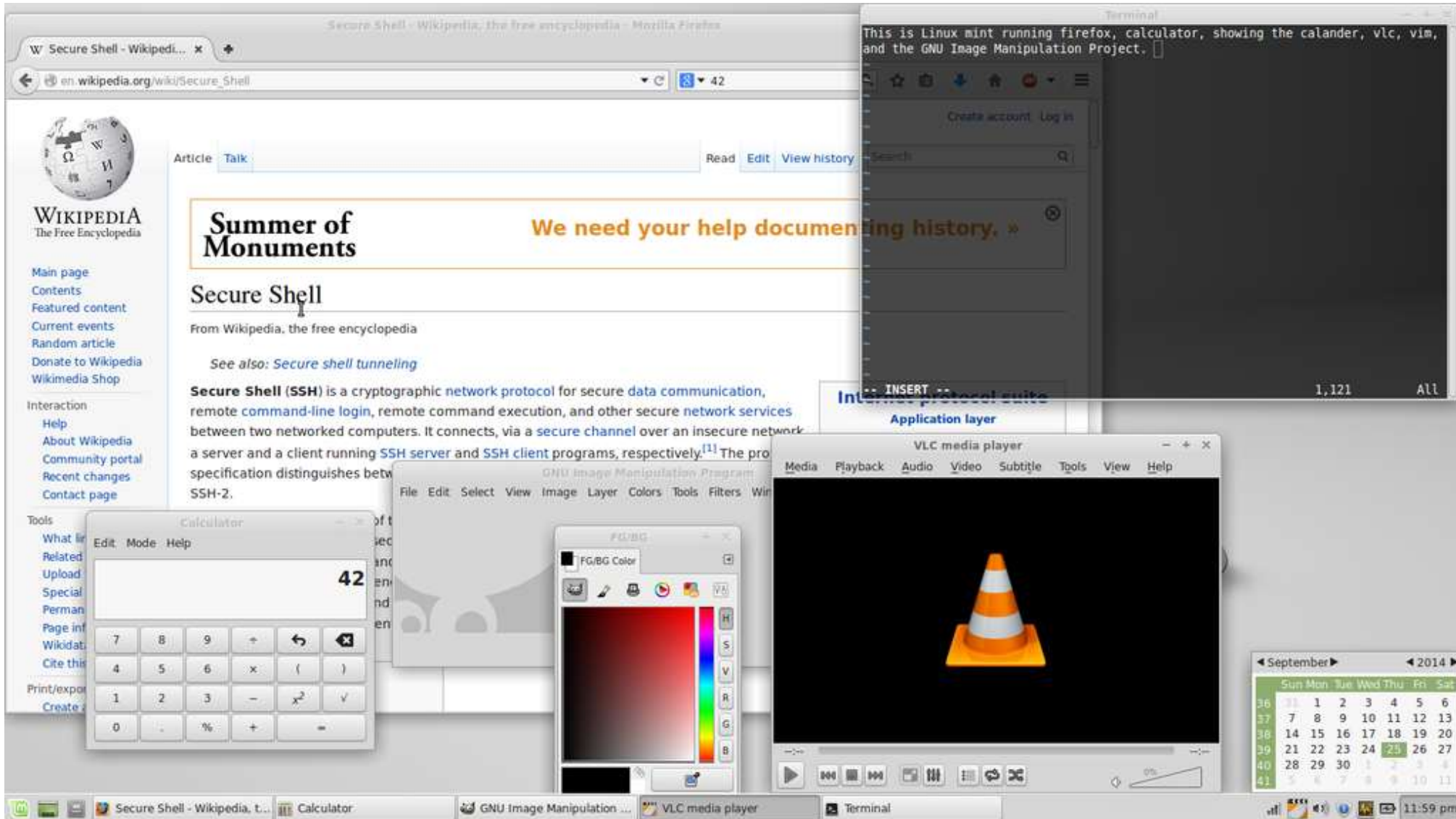
- **New versions of UNIX are designed to use more advanced hardware.**
- Given proper hardware support,
 - OS may be broken into pieces
 - that are smaller and more appropriate than those allowed
 - by the original MS-DOS or UNIX systems.

UNIX

- Unix is a family of multitasking, multiuser computer operating systems that derive from the original AT&T Unix,
- Development starting in the **1970s at the Bell Labs research center by Ken Thompson, Dennis Ritchie, and others**

MULTITASKING

Multitasking is the concurrent execution of multiple tasks (also known as processes) over a certain period of time. New tasks can interrupt already started ones before they finish



MULTIUSER

MULTIUSER

Some multi-user operating systems such as Windows versions from the Windows NT family support simultaneous access by multiple users .

For example, via Remote Desktop Connection as well as the Local user

An example is a Unix or Unix-like system where multiple remote users have access (such as via a serial port or Secure Shell) to the Unix shell prompt at the same time.

UNIX

- There are many different versions of UNIX, although they share common similarities.
- **The most popular varieties of UNIX are Sun Solaris, GNU/Linux, and MacOS X.**

- Some people think Unix and Linux as synonyms, but that's not true.

- **Linux is the clone of Unix. It has several features similar to Unix, still have some key differences.**
- **Linux is a family of open-source Unix-like operating systems based on the Linux kernel**

Difference between Linux and Unix??

Difference between Linux and Unix

Comparison	Linux	Unix
Definition	It is an open-source operating system which is <i>freely available to everyone</i> .	It is an operating system which <i>can be only used by its copyrighters</i> .
Examples	It has different distros like Ubuntu, Redhat, Fedora, etc	IBM AIX, HP-UX and Sun Solaris.
Users	Nowadays, Linux is in great demand. Anyone can use Linux whether a home user, developer or a student.	It was developed mainly for servers, workstations and mainframes.
Usage	Linux is used everywhere from servers, PC, smartphones, tablets to mainframes and supercomputers.	It is used in servers, workstations and PCs.
Cost	Linux is <i>freely distributed,downloaded</i> , and distributed through magazines also. And priced distros of Linux are also cheaper than Windows.	Unix copyright vendors decide different costs for their respective Unix Operating systems.
Development	As it is open source, it is developed by sharing and <i>collaboration of codes</i> by world-wide developers.	Unix was developed <i>by AT&T Labs, various commercial vendors and non-profit organizations</i> .
Manufacturer	Linux kernel is developed by the <i>community of developers</i> from different parts of the world. Although the father of Linux, Linus Torvalds oversees things.	Unix has three distributions IBM AIX, HP-UX and Sun Solaris. Apple also uses Unix to make OSX operating system.

GUI	Linux is command based but some distros provide GUI based Linux. Gnome and KDE are mostly used GUI.	Initially it was command based OS, but later Common Desktop Environment was created. Most Unix distributions use Gnome.
Interface	The default interface is BASH (Bourne Again SHell). But some distros have developed their own interfaces.	It originally used <u>Bourne shell</u> . But is also compatible with other GUIs.
File system support	<u>Linux</u> supports more file system than Unix.	It also supports file system but lesser than Linux.
Coding	Linux is a Unix clone, behaves like Unix but doesn't contain its code.	Unix contain a completely different coding developed by AT&T Labs.
Operating system	Linux is just the kernel.	Unix is a complete package of Operating system.
Security	It provides <u>higher security</u> . <u>Linux</u> has about 60-100 viruses listed till date.	Unix is also highly secured. It has about 85-120 viruses listed till date
Error detection and solution	As Linux is open-source, whenever a user post any kind of threat, developers from all over <u>the world start working on it</u> . And hence, it provides faster solution.	In Unix, users have to wait for some time <u>for the problem to be resolved</u> .